Q1 Report: Used Car Dataset Analysis

Executive Summary

This report presents the data cleaning, transformation, and analysis performed on the used car dataset (train.csv). The primary objective was to convert raw, inconsistent real-world data into a clean, structured format suitable for analysis and machine learning applications. After processing, the dataset was analyzed to compare car brands based on price, power, and usage patterns. The results show that luxury brands such as BMW, Mercedes-Benz, and Audi tend to have higher average prices and power, while brands like Toyota and Mahindra exhibit higher annual usage.

1. Data Cleaning and Validation

The dataset consists of 5,847 rows and 14 columns. Several columns such as Mileage, Engine, Power, and New_Price were stored as text with units included, making them unsuitable for direct numerical analysis.

```
📊 Shape: (5847, 14)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5847 entries, 0 to 5846
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         5847 non-null   int64
 1   Name               5847 non-null   object
 2   Location           5847 non-null   object
 3   Year               5847 non-null   int64
 4   Kilometers_Driven  5847 non-null   int64
 5   Fuel_Type          5847 non-null   object
 6   Transmission       5847 non-null   object
 7   Owner_Type         5847 non-null   object
 8   Mileage            5845 non-null   object
 9   Engine             5811 non-null   object
 10  Power              5811 non-null   object
 11  Seats              5809 non-null   float64
 12  New_Price          815 non-null    object
 13  Price              5847 non-null   float64
dtypes: float64(2), int64(3), object(9)
memory usage: 639.6+ KB
```

	Unnamed: 0	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price
0	1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67 kmpl	1582 CC	126.2 bhp	5.0	NaN	12.50
1	2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	13 km/kg	1199 CC	88.7 bhp	5.0	8.61 Lakh	4.50
2	3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77 kmpl	1248 CC	88.76 bhp	7.0	NaN	6.00
3	4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	2013	40670	Diesel	Automatic	Second	15.2 kmpl	1968 CC	140.8 bhp	5.0	NaN	17.74
4	6	Nissan Micra Diesel	Jaipur	2013	86999	Diesel	Manual	First	23.08	1461	63.1	5.0	NaN	3.50

Key Cleaning Actions

- Unit Standardization:

Mileage values were converted to numeric values. 'km/kg' values were converted to kmpl using a factor of 1.6.

Engine values had 'CC' removed.
Power values had 'bhp' removed.
New_Price values were standardized from 'Cr' to 'Lakh'.
- Outlier Correction:
An extreme outlier in Kilometers_Driven (6,500,000 km) was corrected to 650,000 km.
- Missing Value Identification:
Significant missing values were identified, especially in New_Price.

Out[22]:

	Unnamed: 0	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price
1	2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	13 km/kg	1199 CC	88.7 bhp	5.0	8.61 Lakh	4.50
24	26	Nissan Micra Diesel XV	Hyderabad	2012	54000	Diesel	Manual	First	16.48 km/kg	1461 CC	63.1 bhp	5.0	NaN	4.25
56	58	Nissan X-Trail SLX AT	Hyderabad	2010	121812	Diesel	Automatic	First	9.49 km/kg	1995 CC	147.6 bhp	5.0	NaN	7.75
186	194	Honda City 1.5 GXI	Ahmedabad	2007	60006	Petrol	Manual	First	0.0 kmpl	NaN	NaN	NaN	NaN	2.95
200	208	Maruti Swift 1.3 VXi	Kolkata	2010	42001	Petrol	Manual	First	16.1 kmpl	NaN	NaN	NaN	NaN	2.11
709	733	Maruti Swift 1.3 VXi	Chennai	2006	97800	Petrol	Manual	Third	16.1 kmpl	NaN	NaN	NaN	NaN	1.75
723	749	Land Rover Range Rover 3.0 D	Mumbai	2008	55001	Diesel	Automatic	Second	0.0 kmpl	NaN	NaN	NaN	NaN	26.50
1253	1294	Honda City 1.3 DX	Delhi	2009	55005	Petrol	Manual	First	12.8 kmpl	NaN	NaN	NaN	NaN	3.20
1284	1327	Maruti Swift 1.3 ZXI	Hyderabad	2015	50295	Petrol	Manual	First	16.1 kmpl	NaN	NaN	NaN	NaN	5.80

2. Data Transformation and Imputation

Missing values were handled carefully to avoid loss of data. Dropping rows was avoided since missing values were spread across different rows.

Imputation Methods

- Mileage, Engine, Power: Filled using median values.
- Seats: Filled using mode.
- New_Price: Filled using grouped median by car model, followed by global median.
- Validation ensured that New_Price was not lower than Price.

3. Feature Engineering

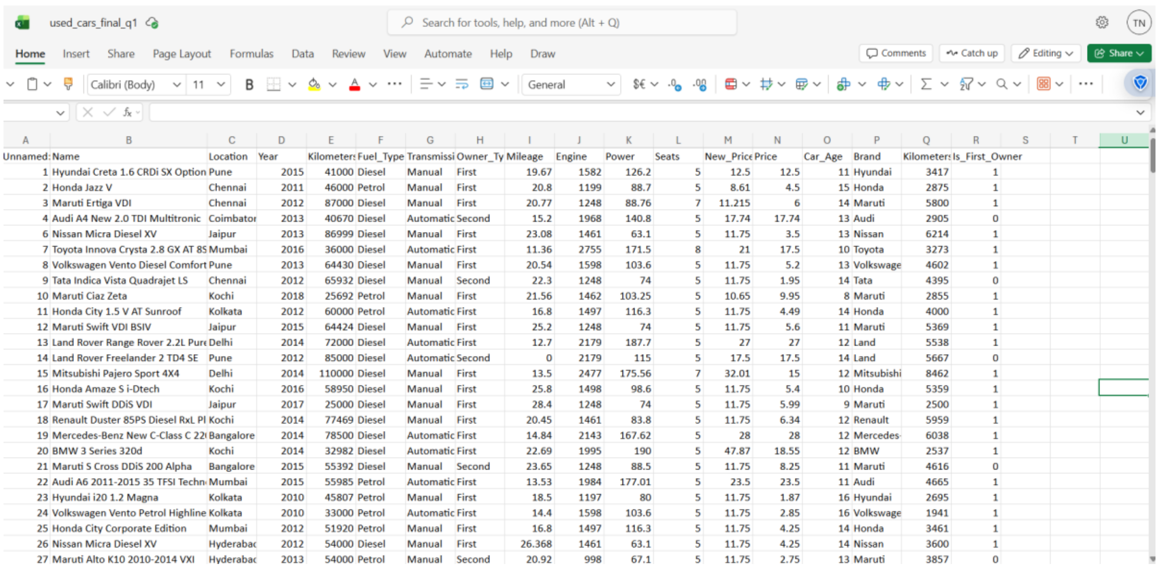
New features were created to enhance the dataset and improve analysis.

- Car_Age: Calculated as 2026 minus the Year of manufacture.
- Brand: Extracted from the first word of the Name column.

- Kilometers_Per_Year: Calculated as Kilometers_Driven divided by (Car_Age + 1).
- Is_First_Owner: Binary feature indicating whether the car is a first-owner vehicle.

4. One-Hot Encoding

Categorical variables such as Fuel_Type and Transmission were converted into numerical format using one-hot encoding. This resulted in additional binary columns and transformed the dataset into a fully numeric format suitable for modeling.



Unnamed:	Name	Location	Year	Kilometers	Fuel_Type	Transmissi	Owner_Ty	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age	Brand	Kilometers	Is_First_Owner
1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67	1582	126.2	5	12.5	12.5	11	Hyundai	3417	1
2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	20.8	1199	88.7	5	8.61	4.5	15	Honda	2875	1
3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77	1248	88.76	7	11.215	6	14	Maruti	5800	1
4	Audi A4 New 2.0 TDI Multitronic	Coimbator	2013	40670	Diesel	Automatic	Second	15.2	1968	140.8	5	17.74	17.74	13	Audi	2905	0
6	Nissan Micra Diesel XV	Jaipur	2013	86999	Diesel	Manual	First	23.08	1461	63.1	5	11.75	3.5	13	Nissan	6214	1
7	Toyota Innova Crysta 2.8 GX AT 8S	Mumbai	2016	36000	Diesel	Automatic	First	11.36	2755	171.5	8	21	17.5	10	Toyota	3273	1
8	Volkswagen Vento Diesel Comfort	Pune	2013	64430	Diesel	Manual	First	20.54	1598	103.6	5	11.75	5.2	13	Volkswage	4602	1
9	Tata Indica Vista Quadrajet LS	Chennai	2012	65932	Diesel	Manual	Second	22.3	1248	74	5	11.75	1.95	14	Tata	4395	0
10	Maruti Ciaz Zeta	Kochi	2018	25692	Petrol	Manual	First	21.56	1462	103.25	5	10.65	9.95	8	Maruti	2855	1
11	Honda City 1.5 V AT Sunroof	Kolkata	2012	60000	Petrol	Automatic	First	16.8	1497	116.3	5	11.75	4.49	14	Honda	4000	1
12	Maruti Swift VDI BSIV	Jaipur	2015	64424	Diesel	Manual	First	25.2	1248	74	5	11.75	5.6	11	Maruti	5369	1
13	Land Rover Range Rover 2.2L Pure	Delhi	2014	72000	Diesel	Automatic	First	12.7	2179	187.7	5	27	27	12	Land	5538	1
14	Land Rover Freelander 2 TD4 SE	Pune	2012	85000	Diesel	Automatic	Second	0	2179	115	5	17.5	17.5	14	Land	5667	0
15	Mitsubishi Pajero Sport 4X4	Delhi	2014	110000	Diesel	Manual	First	13.5	2477	175.56	7	32.01	15	12	Mitsubishi	8462	1
16	Honda Amaze S i-Dtech	Kochi	2016	58950	Diesel	Manual	First	25.8	1498	98.6	5	11.75	5.4	10	Honda	5359	1
17	Maruti Swift DDiS VDI	Jaipur	2017	25000	Diesel	Manual	First	28.4	1248	74	5	11.75	5.99	9	Maruti	2500	1
18	Renault Duster 85PS Diesel RxL Pl	Kochi	2014	77469	Diesel	Manual	First	20.45	1461	83.8	5	11.75	6.34	12	Renault	5959	1
19	Mercedes-Benz New C-Class C 22(	Bangalore	2014	78500	Diesel	Automatic	First	14.84	2143	167.62	5	28	28	12	Mercedes-	6038	1
20	BMW 3 Series 320d	Kochi	2014	32982	Diesel	Automatic	First	22.69	1995	190	5	47.87	18.55	12	BMW	2537	1
21	Maruti S Cross DDiS 200 Alpha	Bangalore	2015	55392	Diesel	Manual	Second	23.65	1248	88.5	5	11.75	8.25	11	Maruti	4616	0
22	Audi A6 2011-2015 35 TFSI Techn	Mumbai	2015	55985	Petrol	Automatic	First	13.53	1984	177.01	5	23.5	23.5	11	Audi	4665	1
23	Hyundai i20 1.2 Magna	Kolkata	2010	45807	Petrol	Manual	First	18.5	1197	80	5	11.75	1.87	16	Hyundai	2695	1
24	Volkswagen Vento Petrol Highline	Kolkata	2010	33000	Petrol	Automatic	First	14.4	1598	103.6	5	11.75	2.85	16	Volkswage	1941	1
25	Honda City Corporate Edition	Mumbai	2012	51920	Petrol	Manual	First	16.8	1497	116.3	5	11.75	4.25	14	Honda	3461	1
26	Nissan Micra Diesel XV	Hyderabac	2012	54000	Diesel	Manual	First	26.368	1461	63.1	5	11.75	4.25	14	Nissan	3600	1
27	Maruti Alto K10 2010-2014 VXI	Hyderabac	2013	54000	Petrol	Manual	Second	20.92	998	67.1	5	11.75	2.75	13	Maruti	3857	0

5. Final Analysis

The final analysis focused on first-owner cars. The dataset was grouped by Brand and summarized to compute average price, power, and kilometers driven per year.

Key Findings

- Most Expensive Brands: BMW, Mercedes-Benz, Audi.
- Most Affordable Brands: Chevrolet, Tata, Nissan.
- Highest Power: BMW, Mercedes-Benz, Audi.
- Highest Usage: Toyota, Mahindra, Skoda.
- Lowest Usage: Hyundai, Honda, Maruti.

Brand	Average_Price_Lakh	Average_Power_BHP	Average_Kilometers_Per_Year	Count	Popular_list
Mercedes-Benz	29.06	192.66	3444.53	260	Yes
BMW	27.56	211.24	4090.77	205	Yes
Audi	26.21	191.74	3947.15	188	Yes
Toyota	12.67	128.8	5648.36	320	Yes
Mahindra	8.53	121.79	4964.91	219	Yes
Skoda	8.21	128.58	4736.31	144	Yes
Ford	7.99	100.33	4304.88	232	Yes
Hyundai	5.83	92.36	3651.15	882	Yes
Renault	5.77	85.5	4127.91	127	Yes
Honda	5.73	106.12	3762.37	499	Yes
Volkswagen	5.37	94.08	4201.04	266	Yes
Maruti	4.88	75.33	3872.08	972	Yes
Nissan	4.81	85.52	4661.96	75	Yes
Tata	4.14	80.81	4605.37	135	Yes
Chevrolet	3.19	91.29	4164.53	88	Yes
Bentley	59.0	552.0	2286.0	1	No
Porsche	49.26	314.21	3666.36	11	No
Land	43.42	188.23	4512.93	42	No

Conclusion

The dataset was successfully cleaned, transformed, and analyzed. The final dataset is structured, consistent, and ready for machine learning applications. The analysis provides valuable insights into pricing, performance, and usage trends in the used car market.